

Architecting Event-Driven **Agentic** **Systems**

Integrating the *Model Context Protocol (MCP)* with Apache Kafka and FastAPI

Agenda

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The Evolution of Backend Engineering
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Choosing the Right Tool for the Job





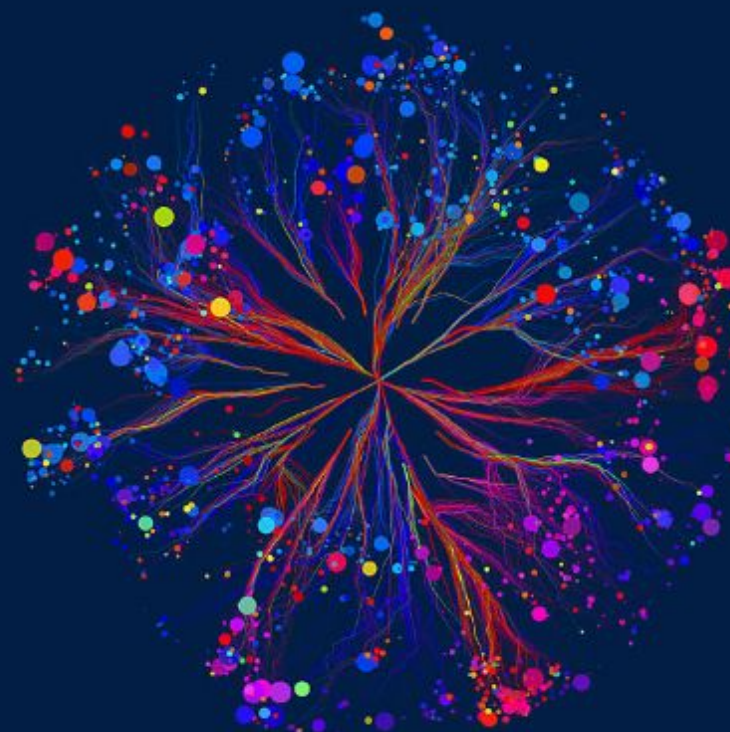
The Paradigm Shift

The Evolution of Backend Engineering

The $N \times M$ Integration Dilemma

Why Traditional Integrations Fail AI

- **Combinatorial Explosion:** Deploying 3 LLMs across 10 internal systems requires 30 custom, bespoke integrations.
- **Brittle Architectures:** Proprietary mechanisms for function calling create fragile, hard-to-maintain glue code.
- **Silent Failures:** AI agents interacting directly with REST APIs often fail due to a lack of runtime tool discovery and context loss.
- **The Result:** Semantic collapse, latency bottlenecks, and a lack of transactional safety.



Resolving Complexity with MCP

The Universal "USB-C" for AI

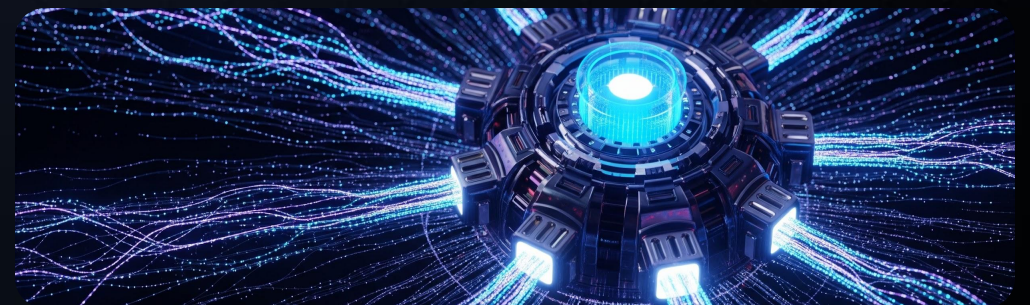
The Model Context Protocol (MCP) collapses the $N \times M$ complexity into a single, standardized interface built on JSON-RPC.

MCP flips the traditional prompt-engineering paradigm. Instead of dumping massive context into a static prompt, models dynamically "pull" exactly what they need.

Any MCP-compliant client can connect securely to any enterprise system without prior developer coordination.

Real-World Examples

- **Claude Desktop:** Fetches real-time GitHub data or local SQLite records without API keys.
- **LangGraph Agents:** Orchestrate multi-step tasks across Google Drive and Slack seamlessly.
- **Enterprise ERP:** Securely query SAP or Oracle records using a standard protocol.



Complementary Layers: REST vs. MCP

Architectural Feature	Traditional REST API Paradigm	Model Context Protocol (MCP)
Primary Consumer	Human-written code (Web apps, microservices)	AI agents and Large Language Models
Design Philosophy	Stateless, discrete operations on defined resources	Stateful, context-preserving orchestration
Interaction Pattern	Call-and-response; Client knows exact endpoint	Exploratory; Agent discovers capabilities at runtime
Documentation	OpenAPI/Swagger optimized for human developers	Natural language designed for LLM intent matching

Agentic Design Patterns



The Reflection Pattern

Mitigates vector similarity flaws by introducing a strict self-review loop. Agents evaluate and refine outputs against enterprise criteria before final transmission, reducing hallucinations.



The Tool-Use Pattern

Operationalized via a centralized Tool Registry. Transitions LLMs from passive text generators to active digital workers with strict input validation and execution guardrails.



The Planning Pattern

Decomposes complex objectives into managed sub-tasks. Utilizes Plan-Act (deterministic upfront planning) or Plan-Act-Reflect-Repeat (dynamic, iterative execution).

Multi-Agent Orchestration

The Router Pattern

Acts as an intelligent API gateway, classifying user intent and delegating to specialized worker agents.

Example: A customer support bot routing billing queries to a finance agent and tech issues to a support agent.

The Sequential Pattern

Agents form a coordinated assembly line where output from one becomes input for the next.

Example: A content pipeline where Agent A extracts text, Agent B translates it, and Agent C summarizes the result.

The Parallel Pattern

Independent sub-tasks are processed concurrently, with results synthesized by a final orchestrator.

Example: Simultaneously searching multiple databases for a person's records to compile a comprehensive profile.

The Competitive Pattern

Multiple agents solve the same problem via different strategies; an evaluator selects the best output.

Example: Three coding agents generate Python functions for a task, and a tester agent selects the one with the least errors.

The Synchronous Bottleneck



Why Traditional Backends Fail Agents

REST APIs are inherently synchronous; an HTTP request expects a near-immediate response. However, an AI agent utilizing a Plan-Act-Reflect loop may require minutes to formulate a plan, execute multiple MCP tool calls, and generate a final synthesis. Attempting to bind complex agentic workflows to synchronous API gateways inevitably results in widespread system failure: timeout cascades, exhausted thread pools, and exceedingly poor resource utilization.

Kafka: The Agentic Nervous System



Out-of-the-Box Asynchrony

Decouples request intake from execution. FastAPI publishes to Kafka and returns HTTP 202 instantly.

- Eliminates client-side timeouts
- Non-blocking AI processing
- Buffers heavy inference spikes



State Preservation & Memory

The persistent log acts as a distributed "source of truth". Agents replay streams for historical context.

- Immutable event sourcing
- Instant context hydration
- Audit trail for agent actions



Horizontal Scalability

Scale worker agents dynamically to handle inference spikes without architectural redesign.

- Parallel LangGraph workers
- Partition-based load balancing
- Dynamic resource allocation



Absolute Resilience (DLQ)

External tool failures do not crash the system. Failed messages route to Dead Letter Queues (DLQ).

- Fault-tolerant tool calls
- Automatic retry logic
- Zero data loss guarantee

Application: Demo & Practice Session

Live Demonstration

Autonomous Edge Incident Response

- ▷ **The Scenario:** Memory leak anomaly detected on remote edge computing device.
- ▷ **Event Trigger:** Telemetry service publishes HighMemoryUsage event to Kafka.
- ▷ **Agentic Reasoning:** LangGraph parses logs via MCP, identifies Docker container fault.
- ▷ **Human-In-The-Loop:** Graph pauses, triggering a Slack webhook for permission to restart.

Interactive Practice Session

Architecting Your First Flow

- ▷ **The Scenario:** Customer Support AI querying an internal billing database for refunds.
- ▷ **Task 1:** Design the @mcp.tool Python signature and identify required guardrails.
- ▷ **Task 2:** Map the LangGraph flow nodes and identify the exact Human-in-the-Loop pause state.
- ▷ **Task 3:** Name the required Kafka topics for event chaining.

Questions?

Thank you

Architecting the Future of Backend Engineering